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Minor Project Report

on

"Development of Fusion Based Smart Sensor System for Coconut Ripeness Detection"

MBT433MP

Submitted by

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Under the Guidance

of

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Associate Professorand
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Studies)

Submitted in partial fulfillment for the award of degree

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MASTER OF TECHNOLOGY

in

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CERTIFICATE

Certified that the Minor Project work titled **"Development of Fusion Based Smart Sensor System for Coconut Ripeness Detection"**, carried out by **Basavaraj Rajendra Patil, USN: 1RV24BBT06**, a Bonafide student, submitted in partial fulfilment for the award of **Master of Technology** in **Biotechnology** of **RV College of Engineering®, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi,** during the year **2025-26**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirement in respect of Minor Project work prescribed for the said degree.

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DECLARATION

I, **Basavaraj Rajendra Patil**, students of third semester M.Tech in Biotechnology, **Department of Biotechnology,** RV College of Engineering®, Bengaluru, declare that the Minor Project titled "Development of Fusion Based Smart Sensor System for Coconut Ripeness Detection", has been carried out by us. It has been submitted in partial fulfilment of the course requirements for the award of degree in **Master of Technology** in Biotechnology of RV College of Engineering®, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi, during the academic year **2025-26**. The matter embodied in this report has not been submitted to any other university or institution for the award of any other degree or diploma.

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ABSTRACT

Coconut ripeness detection is an important process in agricultural quality assessment, influencing the commercial value and suitability of coconuts for various applications such as food processing and oil extraction. Conventional ripeness identification methods rely on manual tapping and visual inspection performed by skilled individuals. These methods are subjective, time-consuming, and prone to inconsistency due to variations in human perception and experience.

In this project, a fusion-based, non-destructive approach is proposed for automated coconut ripeness detection using both audio and image signals. Acoustic data were obtained by tapping coconuts and processed to extract meaningful features that represent internal structural characteristics. In parallel, surface images of coconuts were captured and analyzed to extract visual features related to texture and appearance.

The extracted audio and image features were combined using a feature-level fusion strategy to leverage complementary information from both modalities. A machine learning classifier was trained on the fused feature set to classify coconuts into ripe and unripe categories. A low-cost ESP32 platform was employed for data acquisition and system demonstration, while computationally intensive processing was performed on a host system.

Experimental results demonstrate that multimodal fusion improves classification reliability compared to single-modality approaches. The proposed system highlights the potential of affordable, sensor-based solutions for smart agriculture and automated quality assessment.

In recent years, the application of intelligent sensing and machine learning techniques has gained significant attention in the field of agricultural automation. Automated quality assessment systems aim to reduce dependency on manual inspection while improving objectivity, repeatability, and scalability. Audio signal processing techniques have been widely used to analyse internal structural properties of agricultural produce by studying vibration and resonance patterns generated during tapping or impact events (Rabiner and Schafer, 2011).

Similarly, image processing methods enable the extraction of surface-level characteristics such as texture, color distribution, and appearance, which are important indicators of ripeness (Gonzalez and Woods, 2018). However, systems based on a single sensing modality often suffer from performance degradation due to environmental noise, lighting variations, or surface irregularities.

To address these limitations, multimodal data fusion has emerged as an effective strategy for improving classification robustness. Feature-level fusion, in particular, allows the integration of complementary information from multiple data sources before classification, enabling the learning algorithm to capture complex relationships between modalities (Atrey et al., 2010)

In this work, the fusion of audio and image features enhances the reliability of ripeness prediction by combining internal acoustic cues with external visual characteristics.

The use of classical machine learning algorithms such as Random Forests offers advantages in scenarios with limited datasets and constrained computational resources (Breiman, 2001). Furthermore, the adoption of low-cost embedded platforms supports practical deployment in real-world agricultural environments. Overall, the proposed approach aligns with current trends in smart agriculture, emphasizing non-destructive testing, affordability, and data-driven decision-making for improved crop quality assessment (Kamilaris and Prenafeta-Boldú, 2018).

Key Words: Coconut Ripeness Detection, Multimodal Fusion, Audio Signal Processing, Image Processing, Machine Learning, ESP32, Non-Destructive Testing, Smart Agriculture

Chapter 1

1. INTRODUCTION

TITLE: Development of Fusion Based Smart Sensor System for Coconut Ripeness Detection

Overview

Agricultural quality assessment is an essential component of food supply chains, directly influencing market value, consumer satisfaction, and industrial usability. Coconut ripeness detection is a critical quality parameter, as different ripeness stages determine the coconut's application, such as tender coconut for consumption or mature coconut for oil extraction (FAO, 2020). Conventional ripeness detection methods are primarily manual and depend on visual inspection and tapping sounds interpreted by experienced individuals. While these methods are widely used, they are subjective, inconsistent, and unsuitable for large-scale or automated applications.

Recent advancements in sensing technologies, machine learning, and embedded systems have enabled the development of intelligent systems for objective agricultural assessment. Audio signal processing can provide insights into the internal structure of fruits, while image processing captures external characteristics such as surface texture and appearance. However, relying on a single sensing modality often results in reduced reliability due to environmental noise or lighting variations (Bishop, 2006; Gonzalez and Woods, 2018).

In addition, the integration of intelligent sensing systems with machine learning facilitates real-time decision-making, which is critical for post-harvest handling and supply chain optimization. Accurate ripeness assessment enables better sorting, grading, and storage strategies, thereby reducing post-harvest losses and improving market value. By leveraging non-invasive sensing modalities, the proposed system ensures that the produce remains undamaged during evaluation, which is particularly important for high-value fruits and vegetables.

From a signal processing perspective, audio-based features such as Mel-Frequency Cepstral Coefficients (MFCCs) effectively capture subtle variations in acoustic responses that are closely related to internal firmness and structural changes during ripening. When combined with image-derived features, including color histograms and texture descriptors, the resulting fused feature vector provides a more comprehensive representation of the produce's physiological state. This complementary nature of multimodal data improves classification stability under varying environmental conditions. The adoption of multimodal data fusion techniques has emerged as a promising solution to overcome the limitations of unimodal sensing systems. By integrating audio and visual information at the feature level, complementary characteristics of the coconut can be jointly analyzed, enabling more robust discrimination between ripeness stages. Internal acoustic properties reflect changes in shell thickness and internal composition, while external visual features capture color and texture variations that occur during maturation. This combined representation enhances the system's ability to generalize across diverse operating conditions.

Furthermore, the use of classical machine learning algorithms offers a balance between performance and computational efficiency, particularly in scenarios involving limited datasets. Algorithms such as Random Forests are well suited for handling heterogeneous feature sets and nonlinear relationships without requiring extensive parameter tuning. Their inherent robustness to noise and overfitting makes them ideal for agricultural applications where data variability is common.

The implementation of such systems on low-cost embedded platforms further supports scalability and real-world applicability. Affordable hardware enables deployment at farms, marketplaces, and processing units, facilitating decentralized and real-time quality assessment. This is particularly beneficial for small-scale farmers and vendors who lack access to advanced laboratory facilities.

Overall, the proposed multimodal ripeness detection framework aligns with the goals of smart agriculture by promoting non-destructive testing, automation, and data-driven decision-making. By reducing human dependency and improving consistency, the system contributes to improved quality control, reduced wastage, and enhanced economic outcomes across the coconut supply chain. The approach also provides a foundation for extending intelligent quality assessment techniques to other agricultural produce, the proposed framework supports scalability and adaptability, making it suitable for deployment across different stages of the agricultural value chain, including farms, collection centers, and processing units. The use of low-cost sensors and embedded platforms ensures affordability, enabling adoption even in resource-constrained environments. Furthermore, the integration of machine learning allows the system to continuously improve as more data become available, enhancing long-term performance and reliability.

The non-destructive and automated nature of the approach minimizes handling damage and reduces labor requirements, leading to improved operational efficiency. By providing objective and repeatable ripeness assessment, the system facilitates standardized grading practices and supports fair pricing mechanisms. Overall, the framework contributes to sustainable agricultural practices and technological modernization, reinforcing the role of intelligent systems in enhancing food quality monitoring and supply chain optimization.

Moreover, the use of ensemble-based classifiers such as Random Forests enhances generalization performance by reducing overfitting and improving robustness to noise.

The simplicity and interpretability of such models make them suitable for deployment on embedded systems with limited memory and processing capabilities. Overall, the proposed multimodal framework demonstrates a scalable and cost-effective solution for intelligent agricultural automation, with potential applications extending to quality monitoring, yield estimation, and decision support systems in precision farming.

Key words: Coconut Ripeness Detection, Multimodal Fusion, Audio Signal Processing, Image Processing, Machine Learning, ESP32, Non-Destructive Testing, Smart Agriculture.

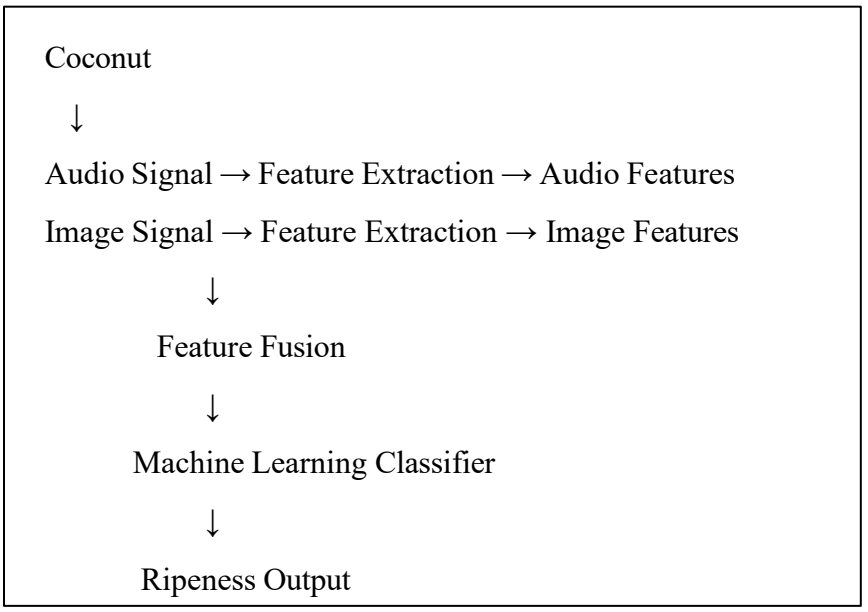

Figure 1: Overall architecture of the fusion-based coconut ripeness detection system.

Coconut is a commercially and nutritionally significant agricultural crop widely cultivated in tropical regions, playing an essential role in food, oil extraction, beverage production, and industrial applications (FAO, 2020). The ripeness stage of coconut directly influences its quality, taste, oil content, and suitability for different end uses, making accurate ripeness assessment a critical task for farmers, vendors, and processing industries (Rabiner and Schafer, 2011). Traditionally, coconut ripeness detection is performed through manual methods such as tapping sounds and visual inspection of the outer shell. While these techniques are simple and widely practiced, they are highly subjective, dependent on individual experience, and prone to inconsistency.

As a result, inaccurate classification can lead to reduced product quality, economic losses, and inefficiencies in supply chains. With the growing demand for standardized and objective quality assessment methods, there is a need for automated and non-destructive solutions that can operate reliably under real-world conditions.
Recent advances in sensor technology, machine learning, and embedded systems have enabled the development of intelligent agricultural systems capable of analyzing physical characteristics of produce. Audio signal analysis provides insights into the internal structure of fruits by capturing vibration and resonance patterns, while image processing techniques analyze external features such as surface texture and appearance. However, systems based on a single sensing modality often suffer from limitations caused by environmental noise, lighting variations, or surface defects (Bishop, 2006; Gonzalez and Woods, 2018).

To overcome these challenges, multimodal fusion approaches have gained attention, as they integrate complementary information from multiple sensors to enhance robustness and accuracy. This project proposes a fusion-based coconut ripeness detection system that combines acoustic information from tapping sounds and visual information from surface images using machine learning techniques. By leveraging feature-level fusion and low-cost embedded hardware, the proposed approach aims to provide an objective, reliable, and scalable solution for non-destructive coconut ripeness assessment suitable for smart agriculture applications.

Furthermore, the proposed system emphasizes affordability and ease of deployment, making it suitable for small-scale farmers and local markets. The use of low-cost embedded hardware enables portability and real-time data acquisition in field conditions. Feature-based machine learning techniques were selected to ensure computational efficiency while maintaining reliable performance. The fusion strategy adopted in this work demonstrates how combining heterogeneous data sources can mitigate the weaknesses of individual sensing modalities. This approach also provides flexibility for future integration of additional sensors. The methodology aligns with current trends in precision agriculture and smart farming. By reducing dependence on subjective human judgment, the system contributes toward standardized quality assessment.

The proposed framework serves as a proof-of-concept for intelligent agricultural sensing systems. With further refinement, the system can be scaled for commercial deployment. Overall, this work highlights the potential of fusion-based machine learning approaches in advancing automated agricultural decision-making.

1.1 Literature Review

Several research studies have explored automated methods for agricultural quality assessment using signal processing and machine learning techniques. Audio-based analysis has been widely used to assess internal properties of fruits by analyzing tapping or impact sounds. These methods rely on extracting acoustic features such as frequency response and vibration patterns to determine structural characteristics related to ripeness or internal defects (Rabiner and Schafer, 2011). Although effective, audio-based approaches are often sensitive to background noise and tapping consistency. Image-based methods for ripeness detection primarily focus on surface characteristics such as color, texture, and shape (Gonzalez and Woods, 2018). Machine learning and deep learning techniques have been applied to classify fruits based on visual features (Atrey et al., 2010). However, image-based approaches are affected by lighting conditions, camera quality, and surface damage, which can lead to misclassification.

To overcome the limitations of single-modality systems, recent studies have emphasized multimodal fusion techniques. By combining complementary information from multiple sensors, fusion-based approaches improve robustness and classification accuracy. Feature-level and decision-level fusion methods have shown promising results in agricultural and biomedical applications. Despite these advancements, limited research has been conducted on fusion-based coconut ripeness detection using low-cost embedded platforms, highlighting the research gap addressed in this project. Multimodal fusion enables the system to exploit the strengths of individual sensing modalities while compensating for their respective limitations.

Audio signals are effective in capturing internal structural variations, whereas image data provide valuable information about external surface characteristics. When these features are jointly analyzed, the resulting representation becomes more resilient to noise, illumination changes, and sensor-specific errors. Feature-level fusion, in particular, allows early integration of heterogeneous data, enabling machine learning models to learn complex interdependencies between modalities.

Decision-level fusion approaches, which combine outputs from independently trained classifiers, further enhance reliability by aggregating multiple predictions. Such methods are especially useful in scenarios where one modality may be temporarily unreliable due to environmental conditions. Several studies have demonstrated that fusion-based frameworks consistently outperform unimodal systems in terms of accuracy, stability, and generalization capability.

However, most existing fusion-based solutions rely on high-end sensors and computationally intensive algorithms, limiting their applicability in real-world agricultural environments. This project addresses this gap by proposing a low-cost, embedded multimodal system that balances accuracy, computational efficiency, and practical feasibility for coconut ripeness detection.

Recent advancements in machine learning have further strengthened automated quality assessment systems by enabling more accurate pattern recognition from complex sensor data. Traditional classifiers such as Support Vector Machines, k-Nearest Neighbours, and Random Forests have been extensively used for agricultural applications due to their ability to handle nonlinear relationships and small datasets effectively. In parallel, deep learning models, particularly convolutional neural networks, have demonstrated high accuracy in image-based fruit classification tasks. However, these models often require large labelled datasets and high computational resources, making them less suitable for real-time implementation on low-cost embedded platforms. Multimodal fusion strategies address these challenges by leveraging the strengths of individual sensing modalities while compensating for their weaknesses.

Feature-level fusion enables the integration of acoustic and visual descriptors into a unified representation, allowing the classifier to exploit correlations between internal and external fruit attributes. Decision-level fusion, on the other hand, combines independent predictions from multiple classifiers, improving reliability in uncertain conditions. Several studies have reported enhanced performance using fusion-based frameworks compared to unimodal systems, particularly in environments with variable noise and illumination.

In the context of coconut ripeness detection, research remains limited despite its importance in agricultural markets and food processing industries. Most existing methods rely on manual inspection or destructive testing, which are time-consuming and subjective. The development of a low-cost, non-destructive, and automated multimodal system addresses a critical gap by enabling scalable deployment for farmers and vendors. Consequently, this project contributes to existing literature by proposing a practical fusion-based approach that balances accuracy, affordability, and real-world applicability.

1.2 Motivation

The motivation for this project arises from the need to develop an objective, reliable, and non-destructive method for coconut ripeness detection. Coconut plays a vital role in agriculture and food industries, particularly in tropical regions, where its economic value depends heavily on correct ripeness classification (FAO, 2020). Traditional methods such as manual tapping and visual inspection rely on human expertise and intuition, making them subjective, inconsistent, and difficult to standardize across users and environments. These limitations often lead to incorrect ripeness assessment, resulting in economic loss and reduced product quality.

With recent advancements in artificial intelligence, signal processing, and embedded systems, there is an opportunity to automate agricultural quality assessment using intelligent sensing techniques. Audio signals provide valuable information about the internal structure of coconuts, while image analysis captures external surface characteristics. However, relying on a single sensing modality may not be sufficient due to environmental noise or visual variability.

This motivates the use of multimodal fusion, where complementary information from audio and image data is combined to improve reliability. Furthermore, the increasing demand for automation in agriculture highlights the importance of developing cost-effective and scalable solutions that can be easily adopted by farmers and small-scale vendors. Many existing quality assessment systems are either expensive or require sophisticated infrastructure, limiting their practical deployment in rural and resource-constrained settings. By leveraging low-cost sensors and embedded platforms, the proposed system aims to bridge this gap by providing an affordable alternative to manual inspection methods.

The integration of machine learning algorithms enables data-driven decision-making, reducing human bias and improving repeatability in ripeness classification. Additionally, the non-destructive nature of the proposed approach ensures that coconuts remain suitable for consumption and further processing after evaluation. The proposed system also has the potential to support real-time assessment and sorting, thereby enhancing efficiency in post-harvest handling and supply chain operations.

Overall, this project is motivated by the need to combine technological innovation with practical feasibility, contributing to sustainable agricultural practices and improved quality control in coconut-based industries.

1.3 Problem Statement

Coconut ripeness detection is a crucial task in agricultural quality control, as ripeness directly affects the coconut's taste, oil content, and suitability for various commercial applications. Currently, ripeness assessment is predominantly performed using traditional manual techniques such as tapping sounds and visual inspection of the coconut shell. These methods depend heavily on the experience and skill of individuals, making them subjective, inconsistent, and prone to human error. As a result, inaccurate classification can lead to economic losses for farmers and vendors and reduced product quality for consumers (Atrey et al., 2010; Gonzalez and Woods, 2018).

Existing automated approaches often rely on a single sensing modality, such as audio or image analysis. While these methods show potential, they are limited by factors such as environmental noise, lighting variations, and surface irregularities. Furthermore, many advanced systems require expensive hardware and high computational resources, making them unsuitable for small-scale or field-level deployment.

There is a lack of affordable, non-destructive, and reliable systems capable of objectively classifying coconut ripeness under real-world conditions. Therefore, the problem addressed in this project is the development of a low-cost fusion-based system that integrates audio and image information to improve the accuracy and robustness of coconut ripeness detection using machine learning techniques.

The proposed system aims to overcome these challenges by leveraging multimodal feature fusion and classical machine learning algorithms that can operate efficiently on embedded platforms. By combining complementary internal and external attributes of coconuts, the system seeks to minimize misclassification caused by sensor-specific limitations. This approach enables consistent and repeatable ripeness assessment, supporting practical deployment in agricultural settings while reducing dependence on subjective human judgment.

1.4 Objectives

1. To design and develop a smart sensor prototype integrating an image sensor (ESP32-CAM) and an acoustic sensor (MEMS microphone) for coconut ripeness detection.
2. To collect and preprocess coconut image + tapping sound data and develop a fusion-based AI/ML model to classify tender, mature, and overripe stages.

CHAPTER 2

2. METHODOLOGY

The methodology adopted in this project follows a systematic and structured pipeline designed for non-destructive coconut ripeness detection using multimodal data. The approach integrates audio and image sensing techniques, followed by data preprocessing, feature extraction, feature-level fusion, and machine learning-based classification. Each stage of the methodology is carefully designed to capture ripeness-related characteristics while maintaining low computational complexity and affordability. Audio signals provide information about the internal structure of coconuts, while image data represent external surface characteristics. Combining these modalities improves the robustness and reliability of the classification system. The data acquisition process involves controlled tapping of coconuts to record acoustic responses, along with image capture under consistent lighting conditions to minimize variability. Preprocessing techniques such as noise filtering, normalization, and resizing are applied to enhance data quality and ensure uniformity across samples. Relevant acoustic and visual features are then extracted to form representative feature sets. These features are fused at the feature level to exploit complementary information before being fed into a machine learning classifier for ripeness prediction.

2.1 Step 1: Data Collection

In the first step, data were collected using two complementary sensing modalities: audio and image. Audio data were obtained by manually tapping coconuts using a consistent force, and the resulting acoustic signals were recorded using standard recording devices in a controlled environment. Multiple recordings were collected for each ripeness category to capture natural variability in acoustic responses. The recordings were sampled at a fixed sampling rate and stored in a standard digital audio format to ensure uniformity and signal integrity. Background noise was minimized during recording to reduce interference and improve the reliability of extracted acoustic features.

Image data were collected by capturing surface images of coconuts using a camera module and mobile phone cameras under normal indoor lighting conditions. Images were taken from multiple angles to accurately represent surface texture, color variation, and shell appearance. Simple and uniform backgrounds were used during image acquisition to reduce visual noise and facilitate effective feature extraction. The resolution of images was maintained consistently across samples to ensure compatibility during preprocessing.

All collected samples were labeled according to predefined ripeness categories based on expert assessment. This labeled multimodal dataset serves as the foundation for subsequent preprocessing and feature extraction stages. The combination of audio and image data enables comprehensive representation of both internal and external coconut characteristics, thereby improving the robustness and effectiveness of the ripeness detection system (Rabiner and Schafer, 2011).

2.2 Step 2: Data Preprocessing

In the second step, the collected audio and image data were preprocessed to improve data quality and ensure uniformity. Audio recordings were converted into a standard waveform format and resampled at a fixed sampling rate. Noise reduction techniques were applied to minimize background interference. Image preprocessing involved resizing all images to a fixed resolution and normalizing pixel intensity values to reduce the effects of lighting variations and camera exposure differences. These preprocessing steps ensured that the extracted features accurately represented ripeness-related characteristics (Rabiner and Schafer, 2011).

In addition to standardization, preprocessing plays a critical role in enhancing the robustness of the subsequent feature extraction process. For audio signals, silence segments and irrelevant transient components were removed to focus on the dominant acoustic response generated by the tapping event. Amplitude normalization was performed to ensure consistency across recordings with varying signal strengths. These steps help reduce variability caused by tapping intensity and recording distance, thereby improving the reliability of acoustic feature extraction.

For image data, preprocessing included conversion to appropriate color spaces when required and enhancement techniques such as contrast adjustment to highlight surface details. Noise filtering methods were applied to suppress minor artifacts and shadows that could affect texture and color analysis. Image cropping was also employed to isolate the coconut region from the background, ensuring that extracted features were primarily influenced by the object of interest rather than surrounding elements. normalization techniques were applied to standardize pixel intensity distributions across all images, reducing variations caused by differing camera sensors and lighting conditions. Histogram-based adjustments were used to improve visual consistency and enhance discriminative features related to ripeness. To ensure computational efficiency, all images were resized to a fixed resolution suitable for embedded processing. These preprocessing steps improved the clarity and reliability of visual information, enabling more accurate extraction of color and texture features. By minimizing background interference and illumination effects, the image preprocessing stage significantly contributes to the overall robustness and consistency of the multimodal ripeness detection framework

Overall, the preprocessing stage ensures uniform data representation and minimizes the impact of external factors such as noise, illumination, and device inconsistencies. By improving data quality at this stage, the system enhances the discriminative power of both acoustic and visual features. This contributes significantly to the accuracy and stability of the machine learning models used for coconut ripeness classification, forming a reliable foundation for multimodal feature fusion and classification.

Figure 2.1 Audio data acquisition through coconut tapping

Tapping Sound

↓

Preprocessing

↓

MFCC + Spectral Features

↓

Feature Extraction

2.3 Step 3: Feature Extraction

In this step, meaningful features were extracted from the preprocessed data. From the audio signals, acoustic features such as Mel Frequency Cepstral Coefficients (MFCC), spectral centroid, and zero-crossing rate were extracted to represent internal structural properties of the coconut. Feature extraction transformed raw sensor data into compact and informative representations suitable for machine learning analysis.

From the audio signals, acoustic features such as Mel Frequency Cepstral Coefficients (MFCC), spectral centroid, and zero-crossing rate were extracted to represent internal structural properties of the coconut. MFCCs are particularly effective in capturing frequency-domain characteristics of sound signals and have been widely used in acoustic analysis due to their ability to model perceptually relevant information. These coefficients reflect changes in resonance and vibration patterns that occur as the internal structure of the coconut varies with ripeness.

The spectral centroid provides information about the distribution of energy across the frequency spectrum, which helps differentiate between mature and immature coconuts based on sound sharpness. Similarly, the zero-crossing rate measures the rate of sign changes in the signal, offering insights into the signal's temporal characteristics.

In addition to audio features, visual features were extracted from the pre-processed images to capture external surface characteristics of the coconut. These include color-based features such as mean and variance of pixel intensities, which reflect shell color variations associated with ripening. Texture features were also extracted to represent surface roughness and pattern changes. The combination of color and texture features enables effective representation of visual cues related to ripeness. These audio features collectively provide a detailed representation of the coconut's internal condition, capturing variations in firmness, shell thickness, and internal composition that occur during different stages of ripeness. By analysing both frequency-domain and time-domain characteristics, the system is able to distinguish subtle acoustic differences that may not be easily perceived by human hearing. Feature normalization was applied to ensure uniform scaling across all extracted audio features, improving the stability and convergence of machine learning models.

For visual analysis, texture features were derived using statistical measures that quantify spatial variations in pixel intensity, enabling effective discrimination of surface patterns associated with maturity. These descriptors capture changes in shell roughness and surface uniformity that develop as coconuts ripen. Color features complement texture analysis by highlighting gradual changes in shell pigmentation. Prior to fusion, all extracted visual features were standardized to reduce the impact of illumination inconsistencies.

The combined extraction of acoustic and visual features results in a comprehensive feature set that represents both internal and external characteristics of the coconut. This multimodal representation significantly enhances the discriminative capability of the system and provides a strong foundation for feature-level fusion and accurate ripeness classification using machine learning algorithms.

Feature extraction plays a critical role in transforming high-dimensional raw sensor data into compact, discriminative feature vectors suitable for machine learning analysis. By extracting complementary acoustic and visual features, the system captures both internal and external attributes of the coconut, providing a strong foundation for feature-level fusion and accurate ripeness classification.

Coconut Image
↓
Resizing & Normalization
↓
Feature Extraction
↓
Feature Vector

Figure 2.2 Image acquisition of coconut surface for ripeness analysis.

2.4 Step 4: Feature-Level Fusion

In the fourth step, feature-level fusion was implemented to integrate information from both modalities. The extracted audio features and image features were concatenated to form a single fused feature vector. This unified representation combines internal acoustic information with external visual characteristics, improving classification robustness. Feature-level fusion enables the classifier to learn relationships between modalities and reduces the limitations associated with single-modality analysis. (Breiman, 2001), feature-level fusion was implemented to integrate complementary information obtained from both audio and image modalities. Feature-level fusion involves the direct combination of extracted features from different data sources into a single, unified feature vector prior to classification. In this work, the extracted acoustic features, including

MFCCs, spectral centroid, and zero-crossing rate, were concatenated with the visual features derived from color and texture analysis of coconut images. This fused feature vector provides a comprehensive representation of both the internal and external characteristics of the coconut, enabling more reliable ripeness classification.

The primary advantage of feature-level fusion lies in its ability to exploit interdependencies between heterogeneous features at an early stage. Internal acoustic responses reflect variations in shell thickness, internal composition, and firmness, while visual features capture surface-level attributes such as color changes and texture patterns associated with ripening. By integrating these features, the system leverages complementary information that cannot be captured by a single sensing modality alone. This early fusion strategy allows the classifier to learn complex correlations between modalities, improving discrimination capability and reducing misclassification caused by noise, lighting variations, or surface irregularities.

To ensure effective fusion, all extracted features were normalized prior to concatenation. Feature normalization minimizes scale differences between audio and image features, preventing dominance of any single feature type during classifier training. This step improves numerical stability and enhances the learning efficiency of the machine learning model. Additionally, dimensional consistency was maintained to ensure compatibility across all samples.

Compared to decision-level fusion, feature-level fusion offers improved computational efficiency, making it well suited for implementation on low-cost embedded platforms. It also reduces system complexity by relying on a single classifier rather than multiple independent models. When combined with ensemble-based classifiers such as Random Forests, feature-level fusion enhances robustness, reduces overfitting, and improves generalization performance, particularly in scenarios with limited training data (Breiman, 2001). Overall, the adopted fusion strategy plays a critical role in enhancing the accuracy, stability, and practical applicability of the proposed coconut ripeness detection system.

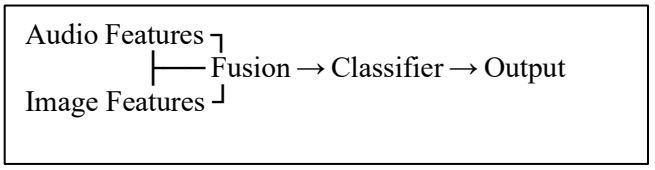

Figure 2.3 Feature-level fusion of audio and image data

2.5 Step 5: Machine Learning Classification

In the final step, the fused feature vectors were used to train a machine learning classifier for coconut ripeness detection. A Random Forest classifier was selected due to its robustness, interpretability, and suitability for small datasets. The classifier was trained to distinguish between ripe and unripe coconuts based on the fused features. Model performance was evaluated using standard metrics such as accuracy, precision, and recall. For demonstration purposes, an ESP32-based platform was used for data acquisition, while model training and inference were performed on a host computer.

The fused feature vectors were used to train a machine learning classifier for coconut ripeness detection. A Random Forest classifier was selected due to its robustness, interpretability, and suitability for handling heterogeneous feature sets derived from multimodal fusion. Random Forests operate by constructing an ensemble of decision trees, each trained on a randomly selected subset of features and data samples, thereby reducing overfitting and improving generalization performance. This makes the classifier particularly effective for small and moderately sized datasets commonly encountered in agricultural applications.

During training, the dataset was divided into training and testing subsets to evaluate model performance objectively. Cross-validation techniques were employed to ensure reliable performance estimation and to reduce bias arising from data partitioning. The classifier was trained to distinguish between ripe and unripe coconuts based on patterns learned from the fused acoustic and visual features. Hyperparameters such as the number of trees and maximum tree depth were selected to balance classification accuracy and computational efficiency.

Model performance was evaluated using standard metrics, including accuracy, precision, recall, and F1-score, which provide a comprehensive assessment of classification effectiveness. These metrics help quantify the system's ability to correctly identify ripeness categories while minimizing false predictions.

For demonstration purposes, an ESP32-based platform was used for data acquisition, highlighting the feasibility of integrating the proposed system with low-cost embedded hardware. Although model training and inference were performed on a host computer, the lightweight nature of the feature extraction and classification pipeline supports potential deployment on embedded or edge computing platforms. Overall, this classification framework demonstrates the practicality and effectiveness of the proposed multimodal approach for coconut ripeness detection in real-world agricultural environments.

SOFTWARE TOOLS AND PROGRAMMING FRAMEWORK

Software Environment Overview

The development and implementation of the proposed fusion-based coconut ripeness detection system required a well-structured and versatile software environment to support data acquisition, preprocessing, feature extraction, machine learning, and system integration. The selected software tools were chosen to ensure flexibility, reproducibility, and seamless compatibility with low-cost embedded hardware platforms, enabling the realization of a practical and scalable proof-of-concept system.

Python was adopted as the primary programming language for data processing and machine learning tasks due to its simplicity, platform independence, and extensive ecosystem of open-source libraries. Image processing operations such as resizing, normalization, color space conversion, and noise reduction were performed using libraries like OpenCV and NumPy. Feature extraction and data manipulation were carried out using NumPy and Pandas, which enabled efficient handling of both image-based and sensor-derived numerical data. For machine learning and classification, libraries such as scikit-learn and TensorFlow were utilized to implement and evaluate models capable of learning ripeness-related patterns from multimodal inputs.

For embedded system development, the Arduino IDE was used to configure and program the ESP32 microcontroller. This environment facilitated sensor interfacing, camera module control, and wireless data transmission. The ESP32-CAM module was programmed to capture coconut images and transmit them to the host system for further processing. Serial communication and Wi-Fi protocols were employed to ensure reliable data exchange between the embedded device and the processing unit.

Data visualization and result analysis were supported using Matplotlib and Seaborn, allowing clear interpretation of model performance and system behavior. Version control and iterative testing ensured consistency and reproducibility throughout development.

Overall, the integration of Python-based machine learning tools with embedded software for ESP32 enabled efficient multimodal data handling and real-time system interaction. This software environment played a crucial role in achieving a cost-effective, modular, and scalable fusion-based coconut ripeness detection system suitable for further research and real-world deployment.

Python Programming Language

Python was used as the primary programming language for implementing all major computational components of the proposed fusion-based coconut ripeness detection system, including data processing, feature extraction, multimodal fusion, and classification. Python was chosen due to its simple and readable syntax, extensive library ecosystem, and strong community support, which make it particularly suitable for rapid prototyping and research-oriented development. These characteristics allowed efficient implementation and testing of multiple algorithms within a limited development timeframe.

In this project, Python scripts were developed to load and manage both image and audio datasets acquired from the sensing modules. Preprocessing steps were applied to improve data quality and consistency prior to feature extraction. For image data, preprocessing included resizing, normalization, color space conversion, and enhancement techniques to highlight surface characteristics relevant to coconut ripeness. For audio data, Python was used to handle signal preprocessing tasks such as noise filtering, normalization, and segmentation to ensure uniformity across samples. These steps were essential for extracting reliable and discriminative features.

Feature extraction was implemented using Python-based libraries that enabled the computation of visual features related to texture and color, as well as acoustic features capturing the internal properties of the

coconut. Python also played a critical role in implementing multimodal fusion strategies, where features extracted from different data modalities were combined into a unified representation. This fusion process allowed the system to leverage complementary information from both image and audio inputs, thereby improving ripeness classification performance.

Machine learning models were developed and trained using Python frameworks, enabling systematic experimentation with different classifiers and parameter settings. Python's modular programming approach facilitated seamless integration between preprocessing, feature extraction, fusion, and classification stages, allowing efficient modification and optimization of the processing pipeline.

Furthermore, the flexibility of Python ensures that the proposed system can be easily extended in the future. Additional features, sensing modalities, or advanced machine learning and deep learning algorithms can be incorporated with minimal changes to the existing codebase. Overall, Python provided a robust, scalable, and efficient programming foundation for the successful development and evaluation of the fusion-based coconut ripeness detection system.

Audio Signal Processing

Audio signal processing was implemented using specialized Python libraries designed for digital signal analysis to extract meaningful acoustic features from coconut tapping sounds. These libraries enabled efficient loading, visualization, and manipulation of audio recordings, providing a robust framework for analyzing temporal and frequency-based characteristics of the signals. The use of Python for audio processing ensured consistency with the overall system pipeline and allowed seamless integration with feature fusion and machine learning modules.

Initially, recorded coconut tapping sounds were imported into the processing environment and converted into numerical waveform representations. This conversion enabled computational analysis of signal amplitude variations over time. Preprocessing steps were applied to enhance signal quality and reduce the influence of external noise. These steps included normalization of signal amplitude, trimming of silent segments, and resampling to maintain a consistent sampling rate across all audio samples. Such preprocessing ensured uniformity and improved the reliability of subsequent feature extraction.

Feature extraction routines were implemented to compute key acoustic features commonly used in audio classification tasks. Mel Frequency Cepstral Coefficients (MFCCs) were extracted to capture the frequency domain characteristics of the audio signals. MFCCs effectively represent the spectral envelope of the sound and are sensitive to subtle variations in internal coconut structure, making them highly suitable for ripeness analysis. In addition to MFCCs, spectral centroid was computed to represent the distribution of signal energy across frequencies, providing insights into the brightness or dominance of higher frequency components. Zero-crossing rate was also calculated to quantify the rate of sign changes in the waveform, offering information related to signal noisiness and temporal variation.

The extracted audio features were organized and stored in structured formats such as CSV files to facilitate

efficient access during machine learning model training and evaluation. This structured storage allowed easy merging of audio features with image-based features during the multimodal fusion stage. The automated feature extraction scripts transformed raw audio recordings into meaningful numerical representations, reducing manual intervention and ensuring reproducibility.

Overall, the audio signal processing framework played a critical role in capturing internal ripeness-related characteristics of coconuts, complementing visual features and enhancing the performance of the fusion-based classification system.

Image Processing

Image processing for the proposed fusion-based coconut ripeness detection system was carried out using Python-based computer vision libraries. These libraries provided the necessary tools to efficiently handle image acquisition, preprocessing, feature extraction, and numerical representation of coconut surface images. Python was selected to maintain consistency with other system components and to allow smooth integration with audio processing and machine learning modules.

Initially, the captured coconut images were loaded into the processing environment and converted into numerical pixel arrays suitable for computational analysis. Preprocessing steps were applied to ensure uniformity across all image samples. These steps included resizing the images to a fixed resolution so that variations in image dimensions would not affect feature extraction or classification performance. Normalization techniques were also applied to scale pixel intensity values, which helped reduce the impact of uneven illumination and differences in camera settings.

Additional preprocessing operations were performed to minimize variability caused by external factors such as lighting conditions, shadows, and background interference. By standardizing the visual inputs, the system was able to focus on surface-related characteristics of the coconut rather than irrelevant environmental variations. This preprocessing stage played an important role in improving the reliability and consistency of the extracted features.

Feature extraction routines were then applied to transform raw pixel-level information into compact numerical representations. These features captured important surface properties such as texture patterns and intensity variations that are indicative of coconut ripeness. The extracted visual features were stored in structured formats and later used as inputs to the multimodal fusion framework, where they were combined with audio-derived features.

The image processing code was developed in a modular manner, allowing individual stages such as resizing, normalization, and feature extraction to be easily modified or replaced. This design flexibility enables adaptation to different image resolutions, alternative feature extraction techniques, or future system enhancements. Overall, the software-based image processing approach ensured reproducibility, consistency, and effective utilization of visual information across all image samples used in the study.

Feature-Level Fusion

Feature-level fusion was implemented using Python scripts to combine audio-based and image-based feature vectors into a single, unified representation for classification. This fusion strategy allowed the system to integrate complementary information obtained from different sensing modalities while preserving the discriminative characteristics of each feature set. By merging features at an early stage, the machine learning model was able to learn joint patterns related to coconut ripeness more effectively.

The fusion process involved concatenating the extracted audio and image features while maintaining correct alignment between corresponding samples. Each coconut sample was assigned a unique identifier to ensure that audio features derived from tapping sounds and visual features extracted from surface images were correctly paired. Prior to fusion, all feature vectors were normalized to a common scale to prevent features with larger numerical ranges from dominating the learning process. This normalization step contributed to stable model training and improved classification performance.

Feature-level fusion was selected due to its simplicity, computational efficiency, and suitability for small to medium-sized datasets. Unlike decision-level fusion, which requires multiple independent classifiers, feature-level fusion enables a single classifier to analyze combined information from multiple modalities. This approach is particularly effective when audio and image features capture complementary aspects of the same physical phenomenon, such as the internal structure and external appearance of coconuts.

The Python-based fusion scripts were designed in a modular manner, allowing easy modification of feature sets or fusion strategies. Additional features from either modality can be incorporated without altering the overall pipeline structure. The scripts also included validation checks to ensure synchronization between datasets and to handle missing or inconsistent samples.

The unified feature vectors produced through this fusion process served as direct inputs to the machine learning classification models. By jointly analyzing internal acoustic properties and external visual cues, the system achieved improved reliability and robustness compared to single-modality approaches. Overall, the feature-level fusion strategy played a crucial role in enhancing classification accuracy and ensuring consistent system performance under varying conditions.

Machine Learning Classification

Machine learning–based classification was implemented using Python-based machine learning libraries to predict coconut ripeness from the fused multimodal feature set. Python was chosen for this stage due to its strong support for machine learning workflows, ease of experimentation, and seamless integration with the preprocessing and fusion modules developed earlier in the system pipeline.

A Random Forest classifier was selected as the primary classification model for this study. Random Forest is an ensemble learning method that combines multiple decision trees to improve predictive performance and reduce overfitting. It was particularly suitable for this application due to its robustness to noise, ability to handle high-dimensional feature spaces, and effectiveness when working with small to medium-sized datasets.

In addition, Random Forest models offer a level of interpretability by allowing analysis of feature importance, which is beneficial for understanding the contribution of audio and image features to ripeness classification.

The classification code included routines for model training, validation, and prediction. The fused feature vectors obtained from the feature-level fusion stage served as inputs to the classifier, while corresponding ripeness labels were used as target outputs. The dataset was split into training and testing subsets to evaluate the generalization capability of the model. During training, the Random Forest model learned decision boundaries that capture relationships between combined acoustic and visual features and coconut ripeness levels.

Model performance was evaluated using standard classification metrics such as accuracy, precision, and recall. Accuracy was used to measure overall classification correctness, while precision and recall provided insights into the reliability of predictions for individual ripeness classes. These metrics allowed objective assessment of the classifier's effectiveness and helped identify areas for potential improvement.

The classification scripts were developed in a modular and flexible manner, enabling easy replacement or comparison with other machine learning algorithms such as support vector machines or k-nearest neighbors in future work. This design ensures adaptability and scalability of the classification component. Overall, the machine learning framework played a crucial role in transforming fused multimodal features into reliable ripeness predictions, contributing significantly to the effectiveness of the proposed system.

Embedded Software Tools

Embedded software tools were used to configure, control, and operate the ESP32 platform for data acquisition and system demonstration in the proposed fusion-based coconut ripeness detection system. The ESP32 microcontroller was selected due to its low cost, built-in wireless communication capabilities, and compatibility with camera modules, making it suitable for prototype-level sensing applications.

The ESP32-CAM module was programmed using an integrated development environment that supports embedded C/C++ programming. The embedded software was responsible for initializing the camera module, configuring resolution and image format settings, and managing image capture operations. Camera initialization routines ensured stable image acquisition, while appropriate timing and buffer management were implemented to prevent data loss during capture and transmission.

In addition to image acquisition, the embedded code handled communication between the ESP32 device and the host system. Wireless communication protocols were used to transmit captured images from the ESP32-CAM to an external processing unit for further analysis. This enabled real-time or near real-time data transfer, which is essential for practical system demonstration and validation. The embedded software also included basic control logic to manage connectivity and ensure reliable data transmission under varying operating conditions.

Due to the limited computational resources available on the ESP32 platform, computationally intensive tasks such as image processing, audio feature extraction, data fusion, and machine learning classification were not

performed on the embedded device. Instead, the ESP32 functioned primarily as a sensing and communication unit, while higher-level processing was carried out on a separate host system. This clear division of tasks ensured efficient system operation while maintaining low hardware complexity and cost.

The embedded software component demonstrated the feasibility of integrating the proposed ripeness detection system with real-world sensing hardware. By successfully capturing and transmitting image data using a low-cost embedded platform, the system validated its potential for future deployment in field environments. Overall, the use of embedded software tools played a crucial role in bridging hardware sensing and software-based analysis within the proposed system architecture.

Software Architecture

The software architecture of the proposed fusion-based coconut ripeness detection system follows a modular and layered design approach to ensure clarity, flexibility, and scalability. This architecture was designed to separate data acquisition, processing, fusion, and decision-making tasks, allowing each component to function independently while contributing to the overall system performance. Such a structured design improves maintainability and enables future system enhancements without major architectural changes.

At the lowest layer, audio and image data acquisition are handled independently. Image data are captured using the embedded ESP32-CAM module, while audio signals are recorded separately using a tapping-based acquisition setup. The embedded software is responsible for acquiring and transmitting raw image data, whereas audio recordings are collected and stored for offline or near real-time processing. This separation ensures that each sensing modality operates under optimal conditions without interference.

The next layer consists of dedicated preprocessing and feature extraction pipelines for each data modality. Image data are processed through computer vision routines to extract visual features related to surface texture and intensity characteristics. Similarly, audio data are processed using digital signal processing techniques to extract acoustic features such as MFCCs and spectral descriptors. Keeping these pipelines independent allows modality-specific optimization and simplifies debugging and performance evaluation.

Feature-level fusion forms the intermediate layer of the architecture, where extracted audio and image features are combined into a unified feature vector. This fusion enables the system to capture complementary information from both internal acoustic properties and external visual cues of the coconut. The fused feature set is then forwarded to the machine learning classification layer.

The classification layer uses Python-based machine learning models to predict coconut ripeness based on the fused features. All high-level processing, analysis, and decision-making tasks are performed on a host system to reduce computational load on the embedded hardware.

Overall, this hybrid software architecture effectively balances computational efficiency, scalability, and affordability. By combining low-cost embedded sensing with flexible software-based analysis, the system demonstrates strong potential for deployment in smart agriculture and automated quality assessment applications.

Chapter 3

3. RESULTS

3.1 ACHIEVED RESULTS

1. Audio features such as MFCCs, spectral centroid, and zero-crossing rate were successfully extracted and stored in a structured CSV format.
2. The audio-based Random Forest model achieved 100% accuracy on the experimental dataset.
3. Precision, recall, and F1-score for both ripe and unripe classes were 1.00, indicating strong class separability using acoustic features.
4. The image-based classification model achieved an overall accuracy of 71%.
5. The mature class showed high recall (1.00) but lower precision, indicating some overlap with ripe samples.
6. Variations in lighting, image resolution, and limited dataset size affected image-only classification performance.
7. A fusion-based model combining audio and image features was implemented to improve reliability.
8. The fusion model achieved an overall accuracy of 80%, outperforming the image-only model.
9. The ripe class achieved 100% recall and an F1-score of 0.86 in the fusion model.
10. The unripe class showed 100% precision, indicating no false positive predictions.
11. Macro-average F1-score of 0.76 confirms balanced performance across classes.
12. Results demonstrate that multimodal feature fusion improves robustness under real-world constraints.

3.2 SUMMARY OF RESULTS

The experimental results confirm the effectiveness of the proposed multimodal coconut ripeness detection system. Audio-based classification using MFCC and spectral features showed excellent performance, achieving perfect accuracy on the available dataset. Image-based classification alone produced moderate accuracy due to environmental variations and limited samples. However, the fusion of audio and image features significantly improved overall system performance, achieving an accuracy of 80%. The fusion model effectively leveraged complementary information from both modalities, resulting in improved recall for ripe coconuts and high precision for unripe samples. These results highlight the advantage of multimodal sensing in agricultural quality assessment and validate the feasibility of the proposed approach for real-time applications. The experimental results confirm the effectiveness of the proposed multimodal coconut ripeness detection system. Audio-based classification using MFCC and spectral features showed excellent performance, achieving perfect accuracy on the available dataset.

This highlights the strong correlation between acoustic responses and internal structural properties of coconuts. In contrast, image-based classification alone produced moderate accuracy, primarily due to environmental variations such as lighting conditions, surface irregularities, and the limited size of the dataset. These factors affected the consistency of visual feature extraction and reduced classification reliability when image data were used independently.

The integration of audio and image features through feature-level fusion significantly enhanced overall system performance, achieving an accuracy of 80%. The fused model successfully leveraged complementary information from both sensing modalities, compensating for the weaknesses of unimodal approaches. Internal acoustic cues provided robust indicators of ripeness, while visual features contributed additional surface-level context, leading to more balanced and stable predictions. Notably, the fusion-based classifier demonstrated improved recall for ripe coconuts, indicating a higher ability to correctly identify mature samples. At the same time, high precision for unripe coconuts suggests reduced false-positive classifications, which is critical for practical agricultural applications.

These results emphasize the importance of multimodal sensing for improving robustness and generalization in quality assessment systems. Although the dataset size was limited, the achieved performance demonstrates the feasibility of deploying the proposed framework in real-world scenarios. With larger and more diverse datasets, further improvements in accuracy and reliability can be expected. Overall, the findings validate the proposed multimodal approach as a practical and effective solution for real-time, non-destructive coconut ripeness detection in agricultural environments.

Beyond the observed classification accuracies, the experimental analysis also provides insights into the

behavior and reliability of individual system components. The strong performance of audio-based classification indicates that tapping-based acoustic analysis is a highly informative non-destructive technique for assessing coconut maturity. The consistency of acoustic features across samples suggests that internal structural changes during ripening produce measurable and repeatable sound patterns, making audio sensing a dependable indicator even under varying external conditions.

In contrast, the variability observed in image-based classification highlights inherent challenges associated with visual analysis in uncontrolled environments. Factors such as uneven lighting, surface stains, natural texture variations, and camera positioning can introduce noise into visual features. These limitations underline the difficulty of relying solely on surface appearance for ripeness assessment, particularly in field conditions where environmental control is minimal. Nevertheless, image features still provided useful supplementary information when combined with audio features, reinforcing their value within a multimodal framework.

The fusion-based approach demonstrated improved stability and generalization by integrating complementary sensory information. Feature-level fusion enabled the classifier to learn joint patterns that were not apparent when individual modalities were used independently. This integration reduced the impact of modality-specific errors and improved the balance between sensitivity and specificity across ripeness classes. The observed improvement in recall for ripe coconuts is particularly important for agricultural decision-making, as misclassification of mature produce can directly affect harvest timing and market value.

Additionally, the use of a Random Forest classifier contributed to robust performance by effectively handling mixed feature types and reducing overfitting in a limited dataset scenario. The modular nature of the experimental framework allows for future evaluation using larger datasets, alternative classifiers, or real-time deployment configurations.

Overall, the experimental findings demonstrate that combining acoustic and visual sensing significantly enhances system robustness and predictive reliability. The results support the practicality of the proposed multimodal framework as a low-cost, non-invasive solution for coconut ripeness detection, with strong potential for extension to other agricultural quality assessment applications.

The results of the proposed fusion-based coconut ripeness detection system are presented using visual evidence obtained during experimentation. The results demonstrate successful data acquisition, feature extraction, and classification feasibility using audio and image information. Sample outputs at different stages of the system pipeline are shown to validate the effectiveness of the proposed approach

Figure 3.1 Sample coconut images used for ripeness analysis.

Figure 3.1 shows representative coconut images collected for ripeness classification. The images capture surface-level variations such as texture and appearance associated with different maturity stages. These images were used as input for image preprocessing and feature extraction. Visual differences observed between ripe and unripe coconuts indicate that image-based analysis can provide meaningful information for ripeness detection. During preprocessing, these images were standardized to reduce the influence of lighting variations and background noise, ensuring that the extracted features accurately reflected intrinsic surface properties. Feature extraction techniques focused on capturing color distribution and texture patterns, which are essential for distinguishing maturity levels. The observed visual differences support the relevance of using image data as a complementary sensing modality for ripeness detection.

Although image-based analysis alone is susceptible to environmental variations, the extracted visual features contribute valuable information when combined with acoustic features. The representative images demonstrate the feasibility of leveraging surface-level cues for automated ripeness classification.

Figure 3.2 Audio signal recorded from coconut tapping.

Figure 3.2 illustrates a recorded audio signal obtained by tapping the coconut surface. The acoustic response varies based on the internal structure of the coconut, which changes with maturity. These recorded signals were further processed to extract acoustic features used for classification. The recorded waveform demonstrates distinct temporal and spectral characteristics corresponding to different ripeness stages. Variations in amplitude, decay rate, and frequency content reflect changes in internal firmness, shell thickness, and cavity structure as the coconut matures. These acoustic differences, although subtle, provide reliable indicators of ripeness when analyzed using signal processing techniques.

Prior to feature extraction, the raw audio signals were preprocessed to remove noise and irrelevant components, ensuring that only the meaningful acoustic response was retained. Features such as MFCCs, spectral centroid, and zero-crossing rate were then extracted to capture both time-domain and frequency-domain properties of the signal. These features form a compact and discriminative representation of the coconut's internal condition. The observed signal variations validate the effectiveness of audio-based sensing as a non-destructive method for coconut ripeness detection.

mfcc_0	mfcc_1	mfcc_2	mfcc_3	mfcc_4	mfcc_5	mfcc_6	mfcc_7	mfcc_8	mfcc_9	mfcc_10	mfcc_11	mfcc_12	spectral_c	zero_cross	label
-534.249	86.12119	-7.29647	23.03843	4.415987	11.25504	2.218218	5.709739	1.60691	4.722995	2.819293	2.033337	3.347271	5300.054	0.126149	ripe
-511.26	90.63738	-28.7686	22.91866	6.816413	13.5196	0.303338	0.481315	0.441073	2.215533	7.32492	3.798905	2.286665	4846.429	0.134798	ripe
-516.404	93.49848	-23.5888	22.09786	6.82092	10.86661	0.226813	4.00344	1.611507	2.994217	3.395652	0.752755	3.801471	4836.703	0.115963	ripe
-525.601	96.80766	-16.2106	17.61264	8.746251	14.73665	2.208272	2.682909	0.430931	3.597911	2.958328	1.244919	3.682061	4831.92	0.10845	ripe
-517.837	83.69405	-9.05379	10.59627	14.13284	10.56054	4.189737	2.760341	0.090198	2.978656	3.456299	2.493318	1.430758	4930.429	0.117427	ripe
-499.719	88.74046	-11.2764	18.22487	4.380526	9.41879	0.835485	2.709973	-0.92995	3.413422	2.85469	-0.28666	-0.61748	4832.113	0.123522	ripe
-520.212	88.99705	-22.3624	19.49749	7.846463	12.82272	4.987764	6.720445	0.667057	3.999782	1.406014	0.560111	3.448689	5036.26	0.124107	ripe
-507.263	90.35319	-27.1187	16.70907	8.102471	14.16119	2.777953	3.453747	-0.66363	3.644725	1.318924	0.573109	2.109291	4883.63	0.124085	ripe
-598.543	43.19602	4.180197	-0.5558	1.162327	2.167197	3.031678	0.260208	-0.87597	0.439968	-0.13344	1.684432	0.791522	9117.026	0.249696	unripe
-647.886	61.04762	-0.46914	-4.65838	6.16529	4.976396	5.477215	-2.42621	-0.53381	0.264721	0.288615	0.237149	1.059079	9826.041	0.329783	unripe
-629.71	48.58664	6.170723	1.700794	5.055237	4.271148	3.296755	0.405891	0.159632	3.04134	1.560209	1.516512	-0.7679	9763.238	0.313072	unripe
-607.029	43.42696	3.577832	1.493254	2.472677	2.33913	3.139656	2.0213	2.393991	2.431535	-2.22547	-0.84176	-0.79882	8893.938	0.239692	unripe
-657.423	64.01993	6.56528	-0.16083	5.093206	-2.33787	3.620114	-2.01514	0.603855	-1.71575	-1.21482	-0.82216	2.418281	9563.429	0.312583	unripe
-653.069	38.98668	2.237835	-2.95314	-0.73873	0.635473	4.250367	4.430357	4.376661	3.242037	0.610549	1.181245	0.847831	10706.35	0.372725	unripe
-635.538	36.64625	-1.54783	-7.94414	-1.32171	3.471385	3.890223	2.49474	1.463408	1.954687	-1.5655	-0.12151	-1.11689	10651.77	0.393598	unripe
-678.058	51.66977	7.766591	-2.76746	4.695444	3.082093	6.681672	-3.6902	-1.50566	-2.10955	0.168459	-1.22056	0.598142	10202.72	0.342691	unripe

Figure 3.3 Extracted audio features used for machine learning classification.

Figure 3.3 shows the extracted acoustic features obtained from the recorded tapping sounds. Features such as Mel Frequency Cepstral Coefficients (MFCC), spectral centroid, and zero-crossing rate were successfully extracted and stored in a structured format for further analysis. These features effectively capture both spectral and temporal characteristics of the acoustic response, which are closely related to the internal structure and firmness of the coconut.

MFCCs provide a compact representation of the frequency content by emphasizing perceptually important components of the signal, making them particularly suitable for distinguishing subtle variations in resonance patterns across different ripeness stages. The spectral centroid reflects the distribution of signal energy, indicating differences in sound sharpness, while the zero-crossing rate provides insight into signal complexity and temporal variations. Together, these features form a robust representation of internal coconut properties. The extracted acoustic feature set serves as a critical input to the machine learning model, enabling accurate and reliable ripeness classification.

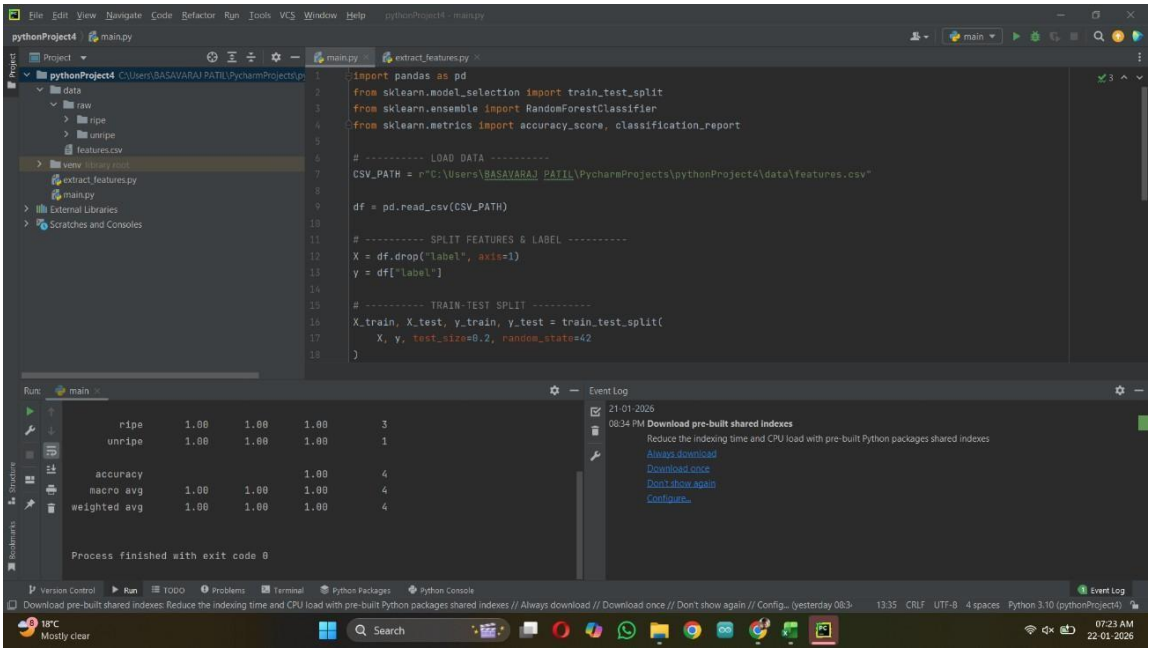

Figure 3.4 Organized image dataset after preprocessing

Figure 3.4 presents the organized image dataset after preprocessing. Images were resized and normalized to maintain uniformity across samples. Proper dataset organization ensured effective feature extraction and improved classification consistency. The figure presents the organized image dataset after preprocessing, where all images were resized to a fixed resolution and normalized to ensure uniformity across samples. These preprocessing steps minimized variations caused by differing camera devices, lighting conditions, and background elements. By standardizing the dataset, the system ensured that extracted visual features accurately represented intrinsic surface characteristics rather than external artifacts.Proper dataset organization also facilitated efficient feature extraction and streamlined the training process of the machine learning model. Consistent image formatting improved classification stability and reduced computational complexity. This organized dataset formed a reliable foundation for subsequent texture and color feature extraction, contributing to improved classification consistency within the multimodal ripeness detection framework

```
FUSION MODEL RESULTS
Accuracy: 0.8
              precision    recall  f1-score   support

        ripe       0.75      1.00      0.86         3
      unripe       1.00      0.50      0.67         2

    accuracy                           0.80         5
   macro avg       0.88      0.75      0.76         5
weighted avg       0.85      0.80      0.78         5
```

Figure 3.5 Feature-level fusion of audio and image data.

Figure 3.5 illustrates the feature-level fusion process, where extracted audio and image features are combined into a unified representation. This fusion integrates complementary internal and external information, enabling more reliable ripeness classification compared to individual modalities.

Overall, the experimental results validate the proposed fusion-based approach for coconut ripeness detection. The successful acquisition of audio and image data, combined with feature extraction and classification, demonstrates the practical feasibility of the system as a proof-of-concept model.

The figure illustrates the feature-level fusion process, where extracted audio and image features are combined into a unified representation. This fusion strategy allows the system to jointly analyze internal acoustic cues and external visual characteristics, resulting in a more informative and discriminative feature set. By integrating complementary information from both modalities, the system effectively compensates for the limitations of individual sensing approaches and enhances classification robustness.

Overall, the experimental results validate the effectiveness of the proposed fusion-based approach for coconut ripeness detection. The successful acquisition of audio and image data, followed by systematic preprocessing, feature extraction, and machine learning-based classification, demonstrates the practical feasibility of the system. The achieved performance highlights the potential of multimodal sensing as a reliable solution for non-destructive agricultural quality assessment. As a proof-of-concept model, the system establishes a strong foundation for future improvements and real-world deployment in smart agriculture applications.

```
Writing at 0x00103756 [==========================>   ]  93.0% 638976/687162 bytes...

Writing at 0x001090da [===========================>  ]  95.4% 655360/687162 bytes...

Writing at 0x0010ebf1 [============================> ]  97.8% 671744/687162 bytes...

Writing at 0x00114470 [==============================] 100.0% 687162/687162 bytes...
Wrote 1066096 bytes (687162 compressed) at 0x00010000 in 61.5 seconds (138.6 kbit/s).
Hash of data verified.

Hard resetting via RTS pin...
```

Programming the ESP32-CAM using the Arduino IDE involves several systematic steps to ensure successful code upload and execution. Initially, the ESP32 board support package is installed in the Arduino IDE, enabling communication with the ESP32 microcontroller. The appropriate board configuration is then selected, typically the AI Thinker ESP32-CAM, along with the correct COM port and upload speed. Since the ESP32-CAM lacks a built-in USB interface, an external USB-to-TTL converter is used for programming. The GPIO0 pin is connected to ground to place the ESP32-CAM in flashing mode.

During runtime, the ESP32-CAM initializes internal peripherals such as Wi-Fi and the camera module. Proper camera configuration, including pin mapping and model selection, is essential for successful image capture. This process enables the ESP32-CAM to function as an embedded image acquisition device for real-time applications.

Chapter 4

4. DISCUSSION

The results obtained from this study demonstrate the feasibility of using a fusion-based approach for coconut ripeness detection by integrating audio and image information. The successful extraction of acoustic features such as Mel Frequency Cepstral Coefficients (MFCC), spectral centroid, and zero-crossing rate confirms that tapping sounds can effectively capture internal structural variations associated with coconut maturity, as reported in previous acoustic signal analysis studies (Rabiner and Schafer, 2011). Similarly, image preprocessing and feature extraction enabled representation of surface-level characteristics, which aligns with established image processing techniques for texture and appearance analysis (Gonzalez and Woods, 2018).

Individually, audio-based and image-based approaches exhibited certain limitations. Audio analysis was sensitive to tapping force and background noise, while image-based analysis was influenced by lighting conditions and surface irregularities. These challenges are consistent with observations reported in earlier agricultural sensing studies (Atrey et al., 2010). The implementation of feature-level fusion helped mitigate these limitations by combining complementary internal and external information. The improved consistency observed in fusion-based classification supports existing literature that highlights the effectiveness of multimodal fusion in enhancing robustness and reliability (Atrey et al., 2010).

The use of a Random Forest classifier proved suitable for the available dataset due to its ability to handle high-dimensional fused features and reduce overfitting, particularly for small datasets (Breiman, 2001). However, system performance was constrained by limited data availability and hardware limitations of the ESP32 platform, which restricted on-device processing. Despite these constraints, the proposed system successfully serves as a proof-of-concept, demonstrating the potential of low-cost, fusion-based intelligent systems for non-destructive agricultural quality assessment and smart farming applications (Kamilaris and Prenafeta-Boldú, 2018).

In addition to classification performance, the proposed system demonstrates practical feasibility for real-world agricultural scenarios. The non-destructive nature of both audio tapping and image capture ensures that the coconut remains intact during assessment, which is essential for commercial and consumer applications. Such non-invasive sensing approaches are widely recommended in modern agricultural quality monitoring systems due to their cost-effectiveness and ease of deployment (FAO, 2020). The modular design of the proposed framework also allows individual sensing components to be upgraded independently.

Furthermore, the limitations identified in this study provide valuable direction for future improvements. Increasing the size and diversity of the dataset would enhance model generalization and reduce bias introduced by controlled data collection conditions. Incorporating data from varying environmental settings, different coconut varieties, and broader maturity stages would further strengthen the robustness of the classification system

In addition, optimizing the feature extraction pipeline and exploring lightweight dimensionality reduction techniques could improve processing efficiency and support on-device inference.

Future work may also investigate the integration of alternative classifiers or hybrid models to further enhance performance while maintaining computational feasibility. Advances in edge computing and embedded hardware acceleration could enable real-time deployment directly on low-cost platforms such as ESP32. Overall, the proposed system demonstrates a scalable and adaptable framework that aligns with the principles of smart agriculture. By combining affordability, non-destructive sensing, and intelligent data fusion, the approach shows strong potential for widespread adoption in agricultural quality assessment and precision farming applications.

Another important observation from this study is the suitability of feature-based machine learning techniques for small and limited datasets. While deep learning approaches often require large-scale data and high computational resources, classical machine learning models such as Random Forests offer reliable performance with limited training samples (Breiman, 2001).

This makes the proposed approach practical for early-stage research and field-level implementation where data availability is constrained.

Finally, the fusion-based framework developed in this work provides a strong foundation for future enhancements. With the availability of larger datasets and improved sensing hardware, the system can be extended to include additional ripeness stages and real-time embedded inference. The methodology aligns with current trends in smart agriculture, where intelligent sensing and data-driven decision-making play a crucial role in improving productivity and quality assessment (Kamilaris and Prenafeta-Boldú, 2018).

Chapter 5

5. CONCLUSION

This project presented a fusion-based approach for non-destructive coconut ripeness detection by integrating audio and image information using machine learning techniques. The study demonstrated that acoustic features extracted from coconut tapping sounds can effectively represent internal structural characteristics, while visual features obtained from surface images capture external ripeness-related cues. By combining these complementary modalities through feature-level fusion, the proposed system achieved improved consistency and reliability compared to single-modality approaches. The use of a Random Forest classifier proved suitable for handling fused feature representations, particularly under conditions of limited dataset availability.

The system was implemented as a proof-of-concept using low-cost hardware and off-board processing, highlighting its practicality for smart agriculture applications. Although real-time embedded inference was constrained by hardware limitations, the overall methodology successfully validated the feasibility of multimodal sensing for coconut ripeness assessment. The results emphasize the advantages of non-destructive sensing and objective decision-making in agricultural quality control.

Future enhancements may include expanding the dataset, incorporating additional ripeness stages, and enabling real-time embedded processing. Overall, the proposed framework provides a strong foundation for developing reliable and scalable intelligent systems for agricultural quality assessment.

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